

joins

=====

practise_joins

create_db.sql

6 tables

show tables

select * from orders;

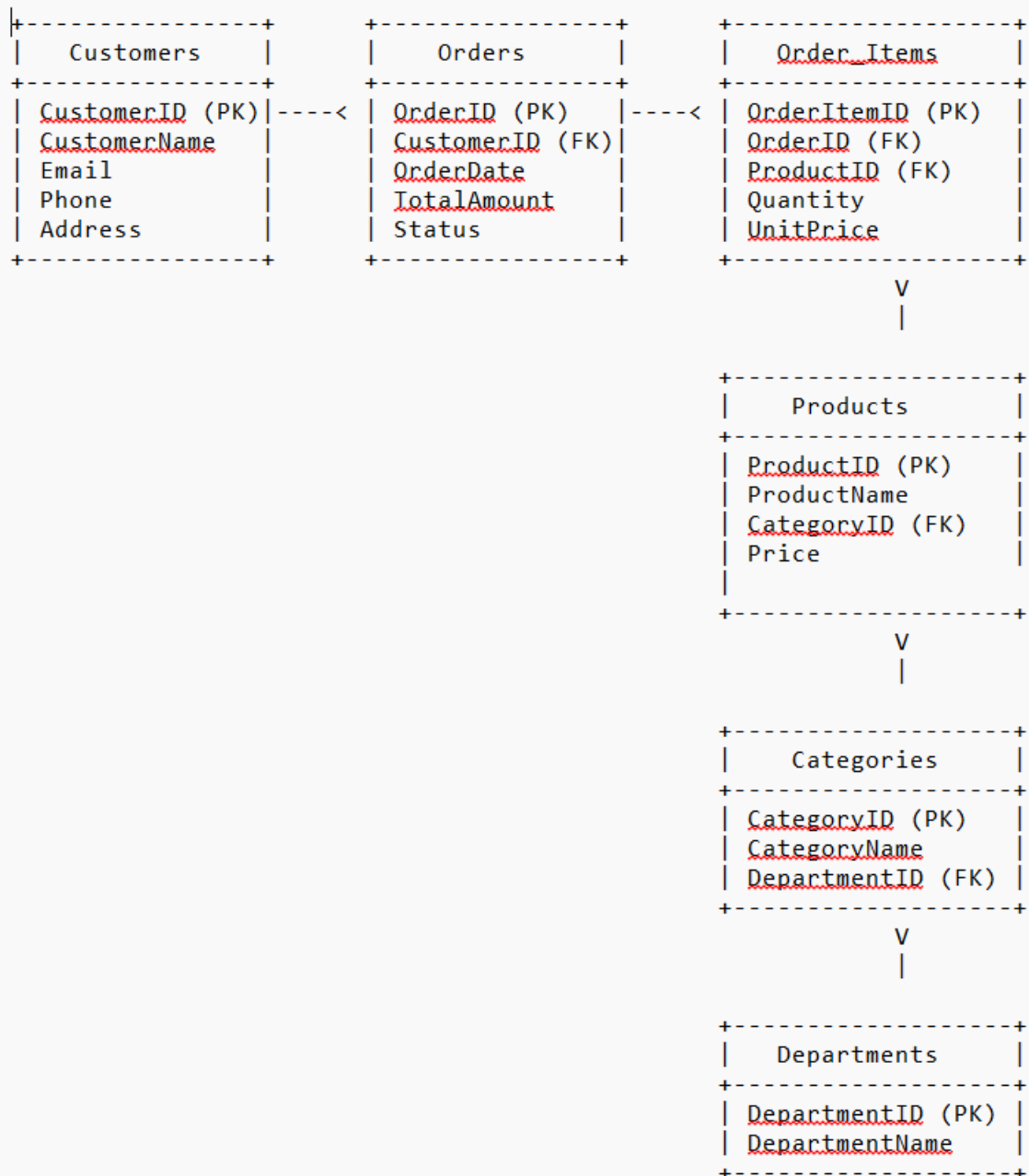
select * from customers;

select * from order_items;

select * from products;

select * from categories;

select * from departments;



What's the name of person who placed the order_id = 1

it can be a 2 step process

step 1 - select order_customer_id from orders where order_id = 1
-- 11599 (customer id)

step 2 –

```
SELECT * FROM customers
WHERE customer_id = 11599;
```

```
SELECT * FROM customers
WHERE customer_id = (select order_customer_id
from orders where order_id = 1);
```

its tedious process

we can join the 2 tables based on common column

orders - order_customer_id
customers - customer_id

names of the columns can be different.

```
SELECT *
FROM orders
JOIN customers
ON orders.order_customer_id = customers.customer_id;
```

```
SELECT *
FROM orders o
JOIN customers c
ON o.order_customer_id = c.customer_id;
```

```
SELECT o.*,c.*
FROM orders o
JOIN customers c
ON o.order_customer_id = c.customer_id;
```

```
SELECT o.order_id , date(o.order_date) , o.order_customer_id ,
o.order_status, c.customer_fname, c.customer_city,
c.customer_state
FROM orders o
JOIN customers c
ON o.order_customer_id = c.customer_id;
```

if both the tables have same column name

orders- customer_id

customers- customer_id

```
SELECT count(*)  
FROM orders o  
JOIN customers c  
ON o.order_customer_id = c.customer_id;
```

-- 68883

what kind of join are we performing?

it is inner join by default

t1 & t2

it will give us the matching records only

100 records 120 records

65 customer ids are matching in both the tables

orders.order_customer_id

2

5

5

5

customers.customer_id

1

2

3

4

5

we will get 4 output records

orders.order_customer_id

2

5

5

5

7

customers.customer_id

1

2

3

4

5

5

2

$$2 + 2 + 2 + 2 = 8$$

It's always a good practise to mention explicitly the type of join.

how many customers have placed atleast one order?

we want records from customers table which have a match in orders table.

101 - 5 orders

101 - 1 customer

101

101

-- 12405 are the number of customers who have placed atleast 1 order..

```
SELECT count(distinct c.customer_id)
FROM orders o
INNER JOIN customers c
ON o.order_customer_id = c.customer_id;
```

```
SELECT customer_id from customers
intersect
SELECT order_customer_id from orders;
```

```
select count(*) from (SELECT customer_id from customers
intersect
SELECT order_customer_id from orders) c ;
```

-- 12405 (customers who placed atleast 1 order)

```
select count(*) from customers;
```

-- 12435 (total customers)

30 customers who have not placed any order.

I want to find customers who are in customers table but not in orders table

except

```
select customer_id from customers
except
select order_customer_id from orders;
```

```
select count(*) from (select customer_id from customers
except
select order_customer_id from orders) c;
```

```
customers who placed atleast 1 order- inner join
intersect
customers who never placed any order- inner join
except
```

